

9 End Term Examination Question Paper

Instructions: -

- i. Attempt all questions and return the question paper with the answer sheet.
- ii. Clearly mention the question number before attempting it

QUESTION 1 [CLO-1, C2, PLO-1]

Explain the following:

- i. Data Cleaning / Cleansing in Data Mining.
- ii. Areas of Application of Data Mining

[10 marks]

QUESTION 2 [CLO-2, C4, PLO-4]

Analyze the given scenario carefully and design ERD with proper labeling and symbols, cardinality and ordinality.

For a scenario on car dealership, the dealership sells both new and used cars, and it operates a service facility. Following are the business rules:

- A salesperson may sell many cars, but each car is sold by only one salesperson.
- A customer may buy many cars, but each car is bought by only one customer.
- A salesperson writes a single invoice for each car he or she sells.
- A customer gets an invoice for each car he or she buys.
- A customer may come in just to have his or her car serviced; that is, a customer need not buy a car to be classified as a customer.
- When a customer takes one or more cars in for repair or service, one service ticket is written for each car.
- The car dealership maintains a service history for each of the cars serviced. The service records are referenced by the car's serial number.
- A car brought in for service can be worked on by many mechanics, and each mechanic may work on many cars.
- A car that is serviced may or may not need parts (e.g., adjusting a carburetor or cleaning a fuel injector nozzle does not require providing new parts).

(Disclaimer: this scenario is taken from Internet resources)

[15 Marks]

QUESTION 3 [CLO-3, C4, PLO-4]

Design Tables for the above scenario (in question #2) and normalize them up to 3NF.

- Mention Primary keys, Foreign Keys properly.
- Do your rough work on last pages.
- Draw only the finalized tables.
- Reverse Engineering is allowed.

Before T1 completes, T2 commits the transaction.
 Alice checks her balance again, expecting \$1000, but now sees \$2000 due to the update from T2.

Q3: [CLO3, C5, PLO4]:

Answer following questions for the given table.

staffNo	dentistName	patientNo	patientName	appointment		surgeryNo
				date	time	
S1011	Tony Smith	P100	Gillian White	12-Aug-03	10.00	S10
S1011	Tony Smith	P105	Jill Bell	13-Aug-03	12.00	S15
S1024	Helen Pearson	P108	Ian MacKay	12-Sept-03	10.00	S10
S1024	Helen Pearson	P108	Ian MacKay	14-Sept-03	10.00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Oct-03	16.30	S15
S1032	Robin Plevin	P110	John Walker	15-Oct-03	18.00	S13

Figure 1: Details of patient dental appointments.

1. Assess above mentioned table and explain any anomalies that may exist? [5]
2. Assess the given and normalize it to 3NF. [10]

Q4: [CLO3, C5, PLO4]:

Assess given schema to identify any dependency(ies), reconstruct the schema to remove the dependency. [5]

Employee(empid, empName, dateOfBirth, specialization)

Salary(sid, empid(FK), date, deductions, allowances, basicSalary, totalSalary)

